

Santiago Cuervo

Speech and Multimodal ML Researcher

✉ santiago.cuervo@lis-lab.fr | 🌐 tiagocuervo.github.io | 🎓 Google Scholar | 🔄 tiagoCuervo

Research Profile

Final-year PhD candidate working to bridge the capability gap between text- and speech-based large language models. My published research spans self-supervised representation learning, deep learning models of human speech perception, scaling laws, and architectures and training algorithms for multimodal adaptation.

Research Experience

Research Scientist Intern *Apple · Machine Learning Research* Mar 2025–Sep 2025

Cupertino, California, USA

Studied the text–speech performance gap in speech-aware LLMs and developed a method to narrow it, producing a model competitive with the strongest open-weight baselines while using over $10\times$ less training data. Published at **ICLR 2026**.

Visiting Student Researcher *Johns Hopkins University · CLSP · JSALT Workshop 2024* Jun 2024–Aug 2024

Baltimore, Maryland, USA

Studied how layerwise representation dynamics govern cross-modal transfer in text-pretrained LMs, and used these insights to develop an architecture which outperforms substantially larger baselines and enables $1.9\times$ faster generation through adaptive depth. Published at **COLM 2026**.

Doctoral Student Researcher *University of Toulon · Aix-Marseille University · CNRS · LIS* Sep 2022–Present

Toulon, France

- Studied scaling behavior, loss–capability relationships, and the effects of data distribution and tokenization in generative speech language models. Showed that their linguistic performance scales up to three orders of magnitude more slowly than that of text language models. Published at **EMNLP 2024**.
- Developed models to predict human speech perception using representations from speech foundation models. This work resulted in the **winning solution to the Clarity Prediction Challenge 2023**, which focused on predicting speech intelligibility for hearing-impaired listeners, as well as publications at **Interspeech 2023 and 2024** and **ICASSP 2024**.

Research Assistant *University of Wrocław* Jul 2021–Jul 2022

Wrocław, Poland

Developed self-supervised algorithms for learning variable-rate speech representations using heuristic segmentation and reinforcement learning, improving downstream speech recognition and enabling the discovery of phone-like units without phonetic or textual supervision. This work resulted in publications at **ICASSP 2022** and **NeurIPS 2022**.

Selected Publications

Closing the Gap Between Text and Speech Understanding in LLMs. Santiago Cuervo, Skyler Seto, Maureen de Seyssel, Richard He Bai, Zijin Gu, Tatiana Likhomanenko, Navdeep Jaitly, and Zakaria Aldeneh. *International Conference on Learning Representations (ICLR)*, 2026.

LF²AR: Accounting for Layerwise Dynamics to Improve Multimodal Adaptation of Language Models. Santiago Cuervo, Adel Moumen, Yanis Labrak, Sameer Khurana, Antoine Laurent, Mickael Rouvier, Phil Woodland, and Ricard Marxer. *Conference on Language Modeling (COLM)*, 2026.

Scaling Properties of Speech Language Models. Santiago Cuervo and Ricard Marxer. *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.

Speech Foundation Models on Intelligibility Prediction for Hearing-Impaired Listeners. Santiago Cuervo and Ricard Marxer. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

Variable-rate Hierarchical CPC Leads to Acoustic Unit Discovery in Speech. Santiago Cuervo, Adrian Łańcucki, Ricard Marxer, Paweł Rychlikowski, and Jan Chorowski. *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Education

PhD in Computer Science <i>University of Toulon, Laboratoire d'Informatique et Systèmes</i> Toulon, France Thesis: <i>Towards Neural Speech Models that Understand.</i> Advisor: Prof. Ricard Marxer.	Sep 2022–Expected Oct 2026
Graduate Studies in Data Science <i>University of Wrocław</i> Wrocław, Poland Coursework completed. Thesis not submitted. GPA: 4.9/5.0; ranked first in a cohort of more than 30 students; Dean's list.	Oct 2020–Jul 2022
Preparatory Course for Studies in Poland <i>Studium Języka Polskiego dla Cudzoziemców, University of Łódź</i> Łódź, Poland	Oct 2019–Jul 2020
Bachelor in Electronic Engineering <i>Universidad Distrital Francisco José de Caldas</i> Bogotá, Colombia Thesis: <i>Emergent Cooperation through Mutual-Information Maximization in a Multi-Agent System.</i> GPA: 4.1/5.0; ranked sixth in a cohort of more than 120 students. Thesis awarded laureate distinction.	Aug 2012–Oct 2019

Technical Skills & Languages

ML LLM pretraining and supervised fine-tuning; speech LLMs; multimodal LLMs; self-supervised representation learning; deep reinforcement learning; interpretability

Infrastructure and Development Python, PyTorch, Linux, SLURM, Weights & Biases; distributed training with DDP, FSDP, and custom parallelism up to 7B parameters / 128 GPUs; training performance profiling; custom research training pipelines; agentic coding workflows with Codex

Languages Spanish (native) | English (C1) | Polish (B1) | French (A2, self-assessed)

Teaching Experience

Graduate Teaching Assistant <i>University of Toulon</i> Toulon, France Master in Marine and Maritime Intelligent Robotics, 60 hours per semester. • Deep Learning: lectures and laboratories on sequence models and self-supervised learning (30 hours). • Data-Driven Machine Perception: supervision of applied deep learning student projects (30 hours).	2022–2024
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Honors and Awards

- Winner, Clarity Prediction Challenge, 2023.
- Ignacy Łukasiewicz NAWA Scholarship, 2019/2020 edition.
- Laureate distinction for bachelor thesis 2020.

Academic Service and Memberships

- Volunteer reviewer for:
 - *NeurIPS 2025.*
 - *ECCV 2026 Workshop on Multimodal Large Language Models for Unified Comprehension and Generation.*
 - *IEEE/ACM Transactions on Audio, Speech, and Language Processing.*
- Member of the Institute of Language, Communication and the Brain (ILCB).